4th Year Project – Health and Safety Risk Assessment and Guidelines.

Every student must perform a health and safety risk assessment of their FYP <u>before</u> commencing any work.

This should be done in consultation with their project supervisor. The FYP Risk Assessment form must be assessed, along with any required material such as Material Safety Data Sheets, and signed off by a member of the Health and Safety Committee.

The form is designed to identify any hazards that the proposed project might present to the student or people that may come into contact with the project.

Issues to consider include

- **Hazardous materials, for example flammable materials, epoxies, cements, paints, aerosols, carcinogens, poisons, toxins…**
- **Working alone, at height, in confined space, in a dangerous area**
- **Working with electricity**
- **Working with machines and tools**
- **Working with bio materials**
- **Working with heat or flame (e.g. soldering or hot air guns)**
- **Working with lasers**
- **Working with heavy or large items**
- **Working with fast spinning objects (e.g. fans or propellers)**
- **Disposable / Recycling of hazardous materials eg batteries, plastics, metal and those mentioned above.**

Where risks are identified, the student must discuss how the risk will be managed by listing what controls are in place and what further controls may be required.

The Risk Assessment Form should be included in your project report.

The Risk Assessment Form should be available for when requested by any staff member.

If this form is NOT available, then all work on the hardware aspects of your project MUST STOP until the form is signed off.

H&S committee members are: Mr. Andrew Meehan, Mr. John Maloco, Ms. Ann Dempsey, and Dr Bryan Hennelly.

I confirm, I have no medical condition that impedes working on my FYP Project, and I will advise a Health & Safety Officer if this status changes.

Signed: Eoin Sheerin

Department of Electronic Engineering, Maynooth University

4th YEAR PROJECT RISK ASSESSMENT FORM Rev. 3

NAME AND STUDENT NUMBER: Eoin Sheerin- 20465956	PROJECT NAME: **FPGA-based Intrusion Detection for DoS Attack**
SUPERVISOR: **Dr. Erivelton Nepomuceno**	PROJECT LOCATION: **E.107**

BRIEF DESCRIPTION OF PROJECT

Using Zc707 board to detect and prevent DOS/DdOS attacks

Hazards, Risk [High(H) Medium (M) Low (L)], and Control Measures

HAZARD	Risk	Controls
Data Privacy Breach	L	Use anonymized or synthetic datasets for training and testing. Ensure no personally identifiable information (PII) is captured or stored. Restrict access to project data.
System Misconfiguration	**L**	Develop a detailed implementation plan and follow strict version control. Use a separate, isolated test network for all experiments to prevent impact on the main network.
Hardware Damage	L	Follow Electrostatic Discharge (ESD) safety procedures (e.g., use of wrist straps). Ensure correct voltage and power supply is used. Operate in a temperature-controlled environment.
Data Loss or Corruption	L	Regularly back up all project code, datasets, and configuration files to a separate, secure location (e.g., cloud storage, external drive). Use version control systems like Git.
Eye Strain from prolonged screen use	l	Ensure proper room lighting and adjust monitor brightness. Take regular breaks.
Electrical Shock when handling the ZC707 board and peripherals	L	Always disconnect the power supply before connecting or disconnecting cables or components. Handle the board by its edges
Mental Fatigue and Burnout	L	**Adhere to a structured work schedule with defined start and end times. Ensure you get adequate sleep and schedule time for hobbies and social activities completely unrelated to the project.**

Identified risks should be discussed with your supervisor and a safe system of work agreed. A more in-depth risk assessment may be required after initial review. Do not proceed until form is signed off.

WEEE Compliance & Recycling: Indicate below how you intend to manage and dispose of your project's material during and post project work.
N/A, project has no material that needs disposing of

Further Controls Required

SIGNATURE OF STUDENT: Eoin Sheerin DATE: 20/10/2025

SIGNATURE OF SUPERVISOR: Erivelton Nepomuceno DATE: 06/11/2025

DEPT HEALTH AND SAFTEY COMMITTEE MEMBER: __ John Mahony __ DATE: __10/12/25